Logic and Scientific Thinking

TECNOLÓGICO DE MONTERREY

Fall 2026

Course Information

Course Number: SO1001
Course Title: Logic and Scientific Thinking
Period: 1
Duration: 5 weeks (August 10 – September 10, 2026)
Schedule: Monday, Wednesday, and Thursday, 8:30 - 10:00 am
Credits: 3

Instructor

Bogdan Popescu
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Office Hours: By appointment

Course Description

This course develops the fundamental skills of critical thinking and logical reasoning. Over five weeks, students learn to reconstruct arguments, detect formal and informal fallacies, see how persuasion and misinformation work, and evaluate evidence and causal claims. The arc runs from *what counts as knowledge* through *argument, validity, and fallacy* to *rhetoric, misinformation, and causation*.

Learning Outcomes

Upon completion of this course, students will be able to:

1. **Reconstruct arguments** in standard form, including the hidden premises a speaker never states.
2. **Name the fallacy** — formal or informal — and run the three-step diagnosis: name the move, quote the words, say what would decide it.
3. **Take apart a persuasive message**: the three appeals in their honest and counterfeit forms, and where misinformation fails — a false premise, a premise stripped of its scope, or the link.
4. **Evaluate evidence**: correlation vs. causation, rival explanations, samples, polls, and source credibility.
5. **Apply the full toolkit** to a real specimen of public persuasion, working from the speaker's own words.

Texts and Resources

See the readings and links under the relevant sessions.

Assessment

The final grade has five components:

- **Midterm exam — 30%**
- **Final exam — 30%**
- **Reading — 20%**
- **Attendance — 10%**
- **Participation — 10%**

Exams

Both exams are closed-book and individual. They test knowledge of the theories and facts developed across the course and independent critical thinking. The **midterm is on Thursday, August 27** and covers Sessions 1–7. The **final exam is on Thursday, September 10** and covers Sessions 10–14. Neither exam is cumulative. The midterm is preceded by a practice-and-review session; before the final, Session 14 runs the whole toolkit on one real study, which serves the same purpose in applied form.

Reading

Each session lists its readings as **Required** or **Optional**. Required readings are done before the session that assigns them; optional ones are there as reference, and nothing rests on them. Reading completion will be tested in the midterm and the final.

Attendance

Attendance is taken every session and is worth 10% of the final grade. See the Attendance policy below for excused absences and the withdrawal threshold.

Participation

Being present is the floor, not the grade. Participation is assessed on contribution to the work of the session — the exercise sheets, team pages, and in-class diagnoses — where the standard is engagement with the argument on the table, not volume of talk.

Course Content

Fifteen 90-minute sessions over five weeks: twelve lecture sessions, a practice-and-review session before the midterm, and the two exams.

Week	Date	Session
1	Mon, Aug 10 Wed, Aug 12	1 · Types of Knowledge 2 · Premises and Conclusions

Week	Date	Session
	Thu, Aug 13	3 · Empirical, Conceptual, and Normative Claims
2	Mon, Aug 17	4 · Deduction and Induction
	Wed, Aug 19	5 · Deductive Tools, Validity, and Truth
	Thu, Aug 20	6 · Formal Fallacies
3	Mon, Aug 24	7 · Informal Fallacies 1
	Wed, Aug 26	8 · Midterm Review — Practice Session
	Thu, Aug 27	9 · Midterm Exam (Sessions 1–7)
4	Mon, Aug 31	10 · Informal Fallacies 2
	Wed, Sep 2	11 · Misinformation
	Thu, Sep 3	12 · Persuasion and Rhetoric
5	Mon, Sep 7	13 · Correlation, Causation
	Wed, Sep 9	14 · Reading a Study Critically
	Thu, Sep 10	15 · Final Exam (Sessions 10–14)

Attendance

Students are required to attend classes following the university's policies. Students with more than three unexcused absences are assumed to have withdrawn from the course. Students with a justified reason for missing class must send an email to the corresponding instructor before the session.

Academic Integrity

The use of artificial intelligence is permitted within the university's ethical guidelines and applicable law, under one condition: every submission includes an appendix stating which AI tools were used, for what purpose, and with which prompts. Undisclosed AI use is academic dishonesty. It is the responsibility of students and instructors, from their respective roles, to know and apply the institutional guidelines.

Any act of academic dishonesty will result in a failing grade on the work in question. Cases of academic dishonesty will be reported to the academic administration.

Students with Special Needs

The university does not discriminate based on disability. Students with approved accommodations must inform their instructors at the beginning of the term. Please see the institutional website for the complete policy.

1 Types of Knowledge

Lecture

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Reading

Sagan, C. (1995). The fine art of baloney detection. In *The demon-haunted world: Science as a candle in the dark* (Ch. 12). Random House. (PDF) **Required.**

2 Premises and Conclusions

Lecture

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Reading

Plato. (n.d.). *Euthyphro*. (PDF) **Required.**

3 Empirical, Conceptual, and Normative Claims

Lecture

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Reading

Hume, D. (1739). *A treatise of human nature*, Book III, Part I, Section I. (PDF) **Required.**

4 Deduction and Induction

Lecture

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Reading

Hume, D. (1748). *An enquiry concerning human understanding*, Section IV, “Sceptical doubts concerning the operations of the understanding.” (PDF) **Required.**

Hempel, C. G. (1966). Scientific inquiry: Invention and test. In *Philosophy of natural science* (Ch. 2). Prentice-Hall. (PDF) **Optional.**

5 Deductive Tools, Validity, and Truth

Lecture

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Reading

Vaughn, L., & MacDonald, C. (2019). *The power of critical thinking* (Canadian ed.). Oxford University Press. (Ch. 3, "Making sense of arguments," pp. 58–106.) (PDF) **Required.**

6 Formal Fallacies

Lecture

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Reading

Kahneman, D. (2011). The characters of the story. In *Thinking, fast and slow* (Ch. 1). Farrar, Straus and Giroux. (PDF) **Required.**

7 Informal Fallacies 1

Lecture

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Reading

Vaughn, L., & MacDonald, C. (2019). *The power of critical thinking* (Canadian ed.). Oxford University Press. (Ch. 5, "Fallacies and persuaders," pp. 151–186.) (PDF) **Required.**

8 Midterm Review

Lecture

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Practice

84 items with answers, covering all seven parts of the paper. Write your answer **and** your reason before opening a box — a correct label with the wrong reason is the most common way to pass practice and fail the exam.

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9 Midterm Exam

Held in class on **Thursday, August 27**.

- Closed book, individual.
- Covers Sessions 1–7: everything up to and including Informal Fallacies 1, **and the required readings assigned for those sessions**.
- Tests knowledge of the theories and facts developed across the course and independent critical thinking.

10 Informal Fallacies 2

Lecture

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Reading

Schopenhauer, A. (c. 1831/1896). *The art of controversy* (T. B. Saunders, Trans.). Swan Sonnenschein. (PDF) **Required.**

11 Misinformation

Lecture

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Reading

Vaughn, L., & MacDonald, C. (2019). *The power of critical thinking* (Canadian ed.). Oxford University Press. (Ch. 4, "Reasons for belief and doubt," pp. 107–150.) (PDF) **Optional.**

Institute for Propaganda Analysis. (1937). How to detect propaganda. *Propaganda Analysis*, *1*(2), 5–8. (PDF) **Required.**

12 Persuasion and Rhetoric

Lecture

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Reading

Aristotle. (n.d.). *Rhetoric*, Book I, Chs. 1–2. (PDF) **Required.**

13 Correlation, Causation

Lecture

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Reading

Vaughn, L., & MacDonald, C. (2019). *The power of critical thinking* (Canadian ed.). Oxford University Press. (Ch. 8, "Inductive reasoning," pp. 260–312.) (PDF) **Optional.**

Huff, D. (1954). The sample with the built-in bias. In *How to lie with statistics* (Ch. 1). W. W. Norton. (PDF) **Required.**

Mill, J. S. (1843). Of the four methods of experimental inquiry. In *A system of logic*, Book III (Ch. 8). John W. Parker. (PDF) **Optional.**

14 Reading a Study Critically

Lecture

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Reading

Dinas, E., Matakos, K., Xefteris, D., & Hangartner, D. (2019). Waking up the golden dawn: Does exposure to the refugee crisis increase support for extreme-right parties? *Political Analysis*, *27*(2), 244–254. (PDF) **Required.**

15 Final Exam

Held in class on **Thursday, September 10**.

- Closed book, individual.
- Covers Sessions 10–14: the second half of the fallacy catalogue, misinformation, persuasion and rhetoric, correlation and causation, and reading a study critically, **and the required readings assigned for those sessions**. Not cumulative.
- Tests knowledge of the theories and facts developed across the course and independent critical thinking.